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MASTER'S INTERNSHIP PROPOSAL · 2027

Decryption failures in lattice-based KEMs: from heuristics to provable bounds

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LOCATION	PQShield, Paris (WeWork Bercy, 8 rue des Pirogues de Bercy, 75012)
DURATION	4–6 months.
TO APPLY	Send a CV, transcripts since L3, a short motivation statement and prior internship reports (if applicable) to damien.stehle@pqshield.com

Context

Most lattice-based Key Exchange Mechanisms (KEMs) and Public-Key Encryptions (PKEs), including the NIST standard ML-KEM [FIPS203], are not perfectly correct: an honestly generated ciphertext fails to decrypt with a small probability δ, typically claimed to be below 2^{-128}. This is not a mere engineering detail: δ appears in the IND-CCA security bound of the Fujisaki–Okamoto transform [HHK17], and decryption failures can be turned into key-recovery attacks [DGJ+19, DRV20].

Yet in nearly all designs, δ is *estimated* rather than *bounded*. Common heuristics include treating the coefficients of noise polynomials as independent, approximating sums by Gaussians, assuming that rounding and compression errors are uniform, and averaging over keys instead of bounding the failure probability of a given key. These shortcuts are not always harmless: dependencies between coefficients can noticeably increase the failure rate of ring and module schemes, in particular when error-correcting codes are used [DVV19].

The NIST PQC process [NIST-PQC], with over twenty lattice-based KEM/PKE submissions across its rounds, and the Chinese NGCC programme [NGCC], whose Round 1 candidates (41 KEMs and 9 key-exchange protocols) were published in September 2026, together provide a huge database of designs and correctness analyses: LWE-, LWR- and NTRU-type schemes over unstructured, ring and module lattices, with binomial, uniform or rounding noise, with or without compression and error correction. Early public analyses of the NGCC candidates already report correctness- and decryption-related flaws [XW26, Saa26].

Objectives

1. **Map the heuristics.** Go through the lattice-based KEMs/PKEs of NIST (all rounds) and NGCC Round 1. For each, record the claimed δ, how it was computed (formula, script, simulation) and every heuristic it relies on.

2. **Identify the failures.** Recompute δ independently, compare with experiments on scaled-down parameters, and pinpoint schemes, parameter sets or keys for which a heuristic is unjustified or the claimed value is wrong. Assess the security consequences.
3. **Prove.** Replace the heuristics by rigorous upper bounds (exact convolutions where feasible, concentration inequalities, careful handling of the dependencies created by ring multiplication, bounds holding for all but a negligible fraction of keys), aiming to lose only a few bits compared to the heuristic estimates for flagship schemes such as ML-KEM.
4. **Tooling** (optional). Package the bounds into an open-source tool that, given a scheme's parameters, outputs a provable upper bound on δ.

Profile

Master's student in cryptography, mathematics or theoretical computer science, with a solid background in probability theory and an interest in lattice-based cryptography. Good programming skills (Python/SageMath; C is a plus).

References

[DGJ+19] J.-P. D'Anvers, Q. Guo, T. Johansson, A. Nilsson, F. Vercauteren, I. Verbauwhede. Decryption failure attacks on IND-CCA secure lattice-based schemes. *PKC 2019*.

[DRV20] J.-P. D'Anvers, M. Rossi, F. Virdia. (One) failure is not an option: bootstrapping the search for failures in lattice-based encryption schemes. *EUROCRYPT 2020*.

[DVV19] J.-P. D'Anvers, F. Vercauteren, I. Verbauwhede. The impact of error dependencies on Ring/Mod-LWE/LWR based schemes. *PQCrypto 2019*.

[FIPS203] NIST. FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard. 2024.

[HHK17] D. Hofheinz, K. Hövelmanns, E. Kiltz. A modular analysis of the Fujisaki–Okamoto transformation. *TCC 2017*.

[NGCC] Institute of Commercial Cryptography Standards (ICCS). Next-generation Commercial Cryptographic Algorithms Program. niccs.org.cn

[NIST-PQC] NIST. Post-Quantum Cryptography Standardization, Round 1–3 submissions. csrc.nist.gov/projects/post-quantum-cryptography

[Saa26] M.-J. O. Saarinen. Chinese NGCC algorithms: the first week of AI cryptanalysis. *IACR ePrint 2026/2266*.

[XW26] Z. Xiong, M. Wang. Cryptanalysis of the ICCS NGCC Round-1 public-key candidates. *IACR ePrint 2026/2232*.